AI-Assisted On-Call Queue: Budgeted Incident Detection for Cloud Microservices

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Abstract

An on-call engineer can read only a few alerts a day, yet most real faults never write a log line, so rules that rank by log severity miss them. This project builds a queue that shows at most K items a day, in the order a live system learns about them, and checks it on the public GAIA microservice benchmark. The queue combines rule-based metric detectors (memory jumps, slow leaks, monitoring gaps) with an optional cost-sensitive logistic-regression classifier for log alerts, and ships as a FastAPI service in one Docker container, in shadow mode by default, with drift monitoring, a shadow challenger, scheduled retraining that rolls back on failure, and a Helm deploy that rolls back on a failed smoke test. On 1,223 test faults at $K = 20$, the detector catches 72.6% (best possible 84.9%) against 68.9% for a Prometheus rate rule and 0% for log severity, and wins all 8 walk-forward weeks with 1.0 instead of 2.1 false items a day. The classifier adds no catches on GAIA, and when the promotion gate is decided on cases that are never scored it refuses the classifier on all six RCAEval applications; plain severity alerts give most of the log-alert gain there. The advantage over the rate rule is modest, every GAIA fault is the same injected memory fault, and no real team has run the system; a replay of real alerts through the loop is not a pilot.

1 Motivation and objectives

Problem. A platform of many small services raises thousands of alerts a day. The engineer on call can read about $K = 20$. Teams escalate by log severity and delete exact duplicates, or use fixed metric thresholds. On GAIA, 82% of the injected faults write no log line and appear only as a jump in container memory, so log severity catches none of them at $K = 20$, and a fixed threshold either misses faults or floods the queue.

Users and value. Users are on-call engineers, the operator who decides when the tool may page, and a team checking it on its own data. The value is a short, ordered, explained list with more real faults and fewer false items.

Objectives. O1: beat the rules a team already runs (log severity, static threshold, Prometheus rate rule) at $K = 20$. O2: decide in real time, using no later data. O3: engineer it for production (authentication, safe paging, versioned models with rollback, monitoring, Docker). O4: test it (at least 90% of tests passing, coverage of at least 85%, p95 latency of at most 1 s at 50 users). O5: let another team evaluate it on its own data without code.

2 Architecture and implementation

The code has two layers. `app/src/` is a library with no HTTP (detectors, causal queue, classifier, evaluation); `app/product/` is the web app, job pipeline, connectors, model registry and monitoring. The queue is first come, first served: a point is hot when it is 0.5 GB above the median

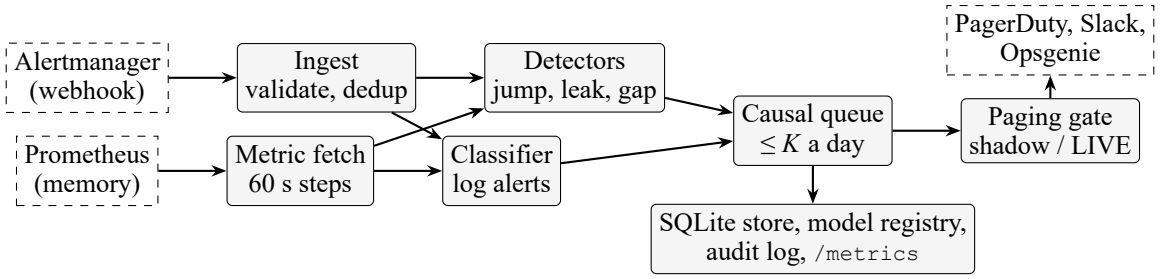


Figure 1: Architecture. Solid boxes run in one container; dashed boxes are external. Engineers' Real/Noise marks flow back through the store into the next model.

of the previous 15 minutes, two hot points within two minutes confirm a jump, and a jump is shown at confirmation only if its rise passes a threshold learned on week 1 (0.62 GB). A 3-hour median catches slow leaks, a service silent for 10 minutes becomes a gap item, and log alerts may use at most a quarter of the slots. Nothing uses a later point, the end of a jump or its peak; `tests/test_causality.py` checks this. Table 1 lists the components and their interfaces.

Component	Responsibility and interface	Main tests
Ingest (`inputs.py`)	Validate and de-duplicate alerts and webhooks; returns `Alert` list	`alertmanager`, `ai_inputs`
Detectors (`memory_spikes.py`)	`detect_jumps/leaks/gaps(metrics, config)` returns `Spike` list	`memory_spikes`
Causal queue (`queue.py`)	`build_queue(items, config)` keeps at most K a day, first come first served	`queue`, `causality`
Classifier, gate (`classifier.py`, `promotion.py`)	Alert probabilities; `promotion_decision(report, incumbent, config)` returns promote or keep	`classifier`, `model_registry`
Paging, store (`paging.py`)	Shadow or LIVE decision per job; SQLite queue, marks, audit log	`would_have_paged`, `security`
MLOps (`drift.py`, `challenger.py`, `retrain.py`)	Input drift (PSI, KS, text classifier test); a refused model tallied against the active one; retrain when due, read back, rolled back on failure	`drift`, `challenger`, `retrain`
Delivery (`helm/`, `deploy.sh`)	One replica, shadow default, password required; atomic install, smoke test, rollback	`helm_chart`, `smoke_script`

Table 1: Components, interfaces and the test files (`app/tests/test_*.py`) that cover them.

Design decisions. (1) Rules do the detection and learning is limited to a confirmation threshold and the log-alert classifier, because a rule can be read and audited. (2) Time split only: train on the first 7 days, test on the next 55, never shuffled. (3) Shadow mode is the default; live paging needs a password, a channel and the typed word LIVE. (4) A new classifier is promoted only if, on held-out data, the queue with it costs at least one missed fault less than the queue without log alerts and the queue it replaces (a miss costs 10, a false item 1); the operator can roll back. (5) Models are stored as skops archives, never pickle. (6) A single worker with SQLite, one container per team. (7) Every model change goes through the loop: a retrain runs only when new marks arrive, passes the gate, is read back from disk and is rolled back if that fails; a refused model keeps being scored in shadow against the active one. **Trade-offs.** Rules are auditable but need a per-team jump size; the strict gate can turn away a useful classifier; shadow mode pages nothing until opted in; SQLite has no failover.

Changes from the proposal. The proposal planned the alert classifier as the main result; on GAIA the metric rules carry it, so the classifier became an optional component behind the promo-

tion gate. A code review removed five uses of later-in-the-day information (full queue 70.2% to 69.6%), and a Prometheus integration test found that a 121 s query step meant live jumps never confirmed; the fetch uses 60 s steps.

3 Testing, evaluation and results

Methods. The suite has 552 tests: unit, API and pipeline integration, AI-specific (malformed rows, a worse model not promoted), security, a Docker test against real Prometheus and Alertmanager and a load test. CI runs lint, types, tests on two Python versions, a dependency audit, a Docker build and the Helm checks, and `run_validate.py` checks every number in the long report. Evaluation uses the 55 test days of GAIA MicroSS (1,223 faults), an 8-week walk-forward, bootstrap intervals and sign tests.

System, $K = 20$ items a day	Caught	False/day	Precision
Log severity + exact duplicates	0.0%	0.8	0%
Static rule (week-1 99th percentile)	57.2%	3.0	81%
Prometheus rate rule	68.9%	2.1	88%
Z-score, EWMA, isolation forest (untuned)	19.7%, 32.4%, 19.8%	14.2, 11.4, 0.3	24%, 39%, 94%
Detector alone (jump + learned threshold)	**72.6%**	**1.0**	94%
Full queue with classifier (refused on GAIA)	69.6%	1.8	90%
Shipped queue (log alerts by severity)	71.0%	1.4	92%

Table 2: Faults caught of 1,223 (best possible at $K = 20$: 84.9%). The detector's 95% bootstrap interval is 70.1–75.1%.

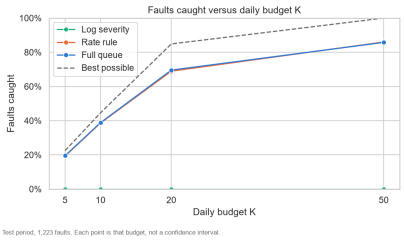


Figure 2: Faults caught versus daily budget K on GAIA. The detector leads at small K and ties the rate rule at 50.

Results. Table 2 shows the detector beats the rate rule in 8 of 8 weeks (sign test $p = 0.0039$) and by 3.7 points overall (interval 2.6 to 4.8), with fewer false items. At $K = 50$ the two are tied (the detector is 0.25 points behind, interval −0.65 to +0.08), so the advantage exists only where slots are scarce. The full queue does not beat it week by week (4 wins, 3 losses, 1 tie). Median time to confirm is 0.72 minutes, and only 2% of misses are the detector's fault. The classifier ablation is the main negative result: with classifier-chosen log alerts the queue catches 69.6%, with severity-chosen alerts 71.0%, with none 71.2%. On RCAEval, an earlier run promoted the classifier on RE2 Train Ticket, but its gate was run on the cases that were scored. I re-ran it with train, validation and test cases kept apart: the gate then refuses the classifier on all six applications (on Train Ticket it saves 4 against a margin of 10). Pooled over 69 RE2 test cases, the shipped queue, which uses severity alerts, catches 68.1% against 60.9% with no log alerts (+7.2 points, interval +1.5 to +14.5), and the classifier adds one case over severity. The app serves 50 users with p95 0.32 s and 0% errors.

Beyond GAIA. Tuned on half of each series, the engine catches 55.1% of 196 SMD incidents

(chance 12.4%), beats chance on NAB traffic and machine data, and catches 0 of 13 RCAEval RE1 memory faults, so the jump size must be set per team.

Production loop. A replay of real alerts with a simulated engineer (not a pilot) ran the GAIA sample (4,852 alerts) and RCAEval Train Ticket (1,050 alerts). On Train Ticket the marks reached the retrain job, which trained 8 times and the gate refused every time. The replay exposed two defects, now fixed: a daily retrain on the same marks (39 refused candidates, now 8) and drift that warned every day. A failing upgrade on a test cluster was rolled back with the data kept. A two-hour soak of the real app, Prometheus and Alertmanager on a simulated fleet (not production) caught 18 of 18 injected faults, no false jump, no failed job; its first run caught 13 of 19, since a second fault soon after another is invisible to the jump rule.

Against the objectives. O1 (beat existing rules): Table 2, 8 of 8 weeks; met for the detector, tied for the full queue. O2 (real time): the causal queue, `test_causality`, median 0.72 min. O3 (production): shadow default, gate, drift, retrain and Helm, tested by `security`, `retrain` and `helm_chart`. O4 (tested): 542 passed, 10 skipped, none failed, 87% coverage, p95 0.32 s. O5 (other teams): `/evaluate` with NAB, SMD and RCAEval. No live pilot was run.

4 Deliverables, setup and use

Repository. `https://github.com/amamsy90/alert-fatigue-reduction`, release 1.1.0 plus the changes in `CHANGELOG.md`, submitted commit: `b9e7424fd6808f4027e5a2fcba28f37f66392809`. The README lists the team, commands, tested system and data sources. No paid API is needed.

Setup and use. From `app/`: `docker compose up -d` serves the app at `127.0.0.1:8080` in shadow mode; `uv sync --group dev`, `uv run pytest -q` and `uv run python run_validate.py` run the tests and check the reported numbers. `uv run python run_reproduce.py` reproduces the GAIA result on a committed 10-day real sample in under a minute; `--full` uses the 1.8 GB data fetched by `tools/fetch_gaia.py`. `tools/deploy.sh` installs the Helm chart, smoke-tests it and rolls back on failure. Workflow: Alertmanager posts alerts, a job builds the day's queue with a reason for each item, the engineer marks items Real or Noise, and a scheduled retrain learns from the marks. The `/evaluate` page runs the same engine on a team's own CSV files.

5 Conclusion

The project delivers a tested, deployable, causal on-call queue with a production MLOps loop. Its measured contribution is a metric detector that catches 72.6% of GAIA faults at 20 items a day with 1.0 false item, against 68.9% and 2.1 for a Prometheus rate rule. Its main findings are negative or limited: the log-alert classifier earns its slots only where faults write logs, the full queue ties the rate rule, and all GAIA faults are one injected shape. The next step is a shadow-mode pilot with a real team.

Limitations and honest use of tools. Solo project; AI coding assistants helped with review, refactoring, tests and documentation (`AGENTS.md`); the running system uses no generative AI; timings are from one Windows laptop.

A Walk-forward results

Test week (2021)	Faults	With metrics	Rate rule	Detector	Full queue
8–15 Jul	161	100%	77.6%	84.5%	82.0%
15–22 Jul	134	91%	73.9%	81.3%	74.6%
22–29 Jul	162	94%	72.2%	77.2%	71.0%
29 Jul–5 Aug	163	82%	67.5%	69.3%	67.5%
5–12 Aug	162	84%	69.1%	71.6%	71.0%
12–19 Aug	139	92%	75.5%	79.1%	74.1%
19–26 Aug	173	61%	51.4%	52.0%	50.9%
26–31 Aug	129	91%	78.3%	83.7%	79.1%
Mean			70.7%	74.8%	71.3%

Table 3: Walk-forward at $K = 20$. The 19–26 Aug fold has a 2.5-day metric outage.

B Classifier on other data

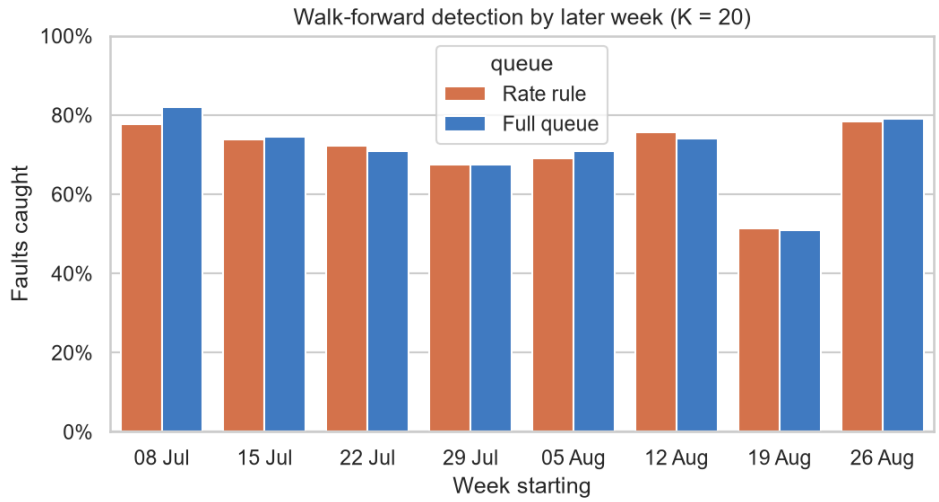
Suite	Application	Cases	No log alerts	Classifier	Gate
RE2	Online Boutique	45	44.4%	46.7%	refused
RE2	Sock Shop	45	84.4%	86.7%	refused
RE2	Train Ticket	45	60.0%	73.3%	promoted
RE3	Online Boutique	15	100%	100%	refused
RE3	Sock Shop	15	93.3%	93.3%	refused
RE3	Train Ticket	15	100%	100%	refused

Table 4: First RCAEval run: share of test faults caught at $K = 10$. The gate ran on the same cases that are scored.

Suite	Application	Cases	No log alerts	Severity	Classifier	Gate (validation)
RE2	Online Boutique	23	43.5%	47.8%	47.8%	refused
RE2	Sock Shop	23	95.7%	95.7%	95.7%	refused
RE2	Train Ticket	23	43.5%	60.9%	65.2%	refused
RE3	Online Boutique	8	100%	100%	100%	refused
RE3	Sock Shop	8	87.5%	87.5%	87.5%	refused
RE3	Train Ticket	8	100%	100%	100%	refused

Table 5: Independent gate test at $K = 10$: train, validation and test cases kept apart; the gate sees only train and validation.

C Walk-forward figure



Each bar is that week's detection rate. The spread is week-to-week variation, not a confidence interval.

Figure 3: Walk-forward detection by later week.